MCEN90054: DESIGN ANDMANUFACTURING PRACTICE-ASSIGNMENT 1

Redesign of 3D printed parts for CNC, Sheet Metal and Injection Moulding.

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SUMMARY

This report details the redesign of three Prusa 3D printer parts for higher volume production using alternative manufacturing processes: CNC machining, injection moulding, and sheet metal forming. Parts chosen for redesign are 201 Y Belt Idler for CNC Machining, 602 LCD support for sheet metal and 908 PSU cover for injection moulding. Key changes include adding fillets, reducing complexities by removing unwanted elements like logos and eliminating all internal Parameters other than the ones necessary for CNC machining, draft angle considerations for injection moulding and redesign of holes, and reducing thickness by complete redesign of part without affecting functionality and adding bending angles for sheet metal bending. Through collaborative team efforts, the final designs achieved manufacturability without compromising performance. Recommendations emphasize leveraging the strengths of each manufacturing process to meet production demands efficiently while maintaining part quality and functionality.

INTRODUCTION

This report documents the redesign of 3D printed parts for CNC machining, injection moulding, and sheet metal fabrication. The chosen parts are the 201 Y Belt Idler for CNC machining, 602 LCD support for sheet metal, and 908 PSU cover for injection moulding. The redesign focused on manufacturability by considering factors like material selection, draft angles, tool accessibility, and bend radius. The aim is to achieve high-volume production while maintaining part functionality and cost-effectiveness.

Q1: CNC PART Design

Part: 201 **Y Belt Idler**

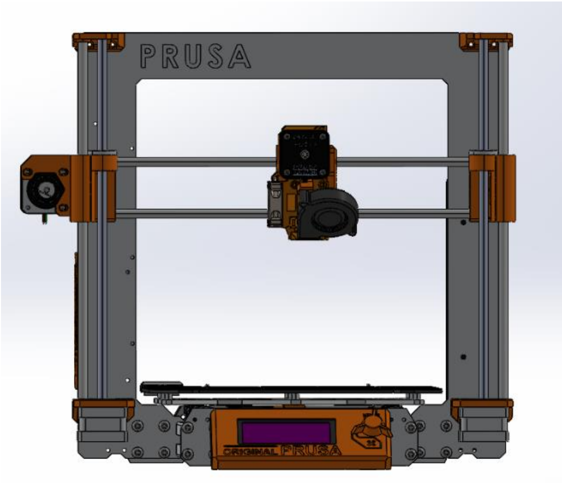


Figure 1. Prusa 3D printer

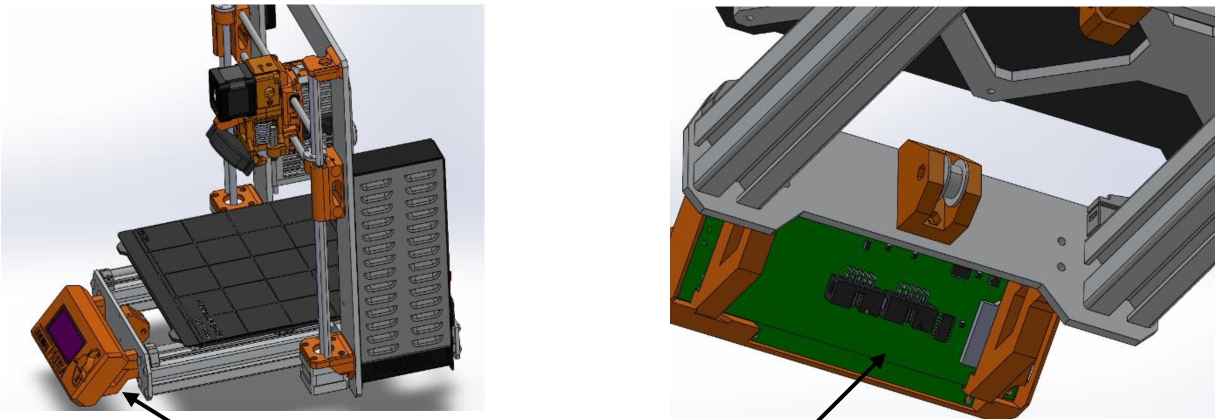


Figure 2 and 3: Y Belt Idler in Prusa 3D

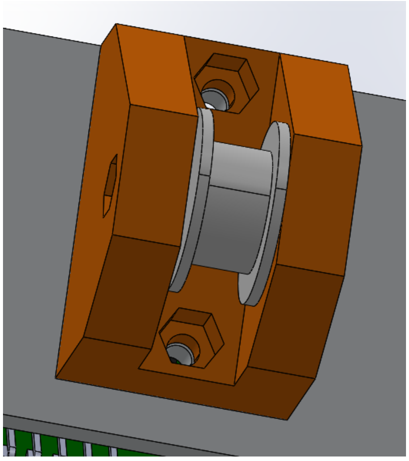


Figure 4: Y Belt Idler fixed with roller fixed on the frame.

Factors to consider:

Material Selection: The current design uses PLA. For CNC machining, we need to select a material that is easily machinable and cost-effective for production volume. Materials like aluminium, mild steel, stainless steel, Titanium, acetal, etc are considered.

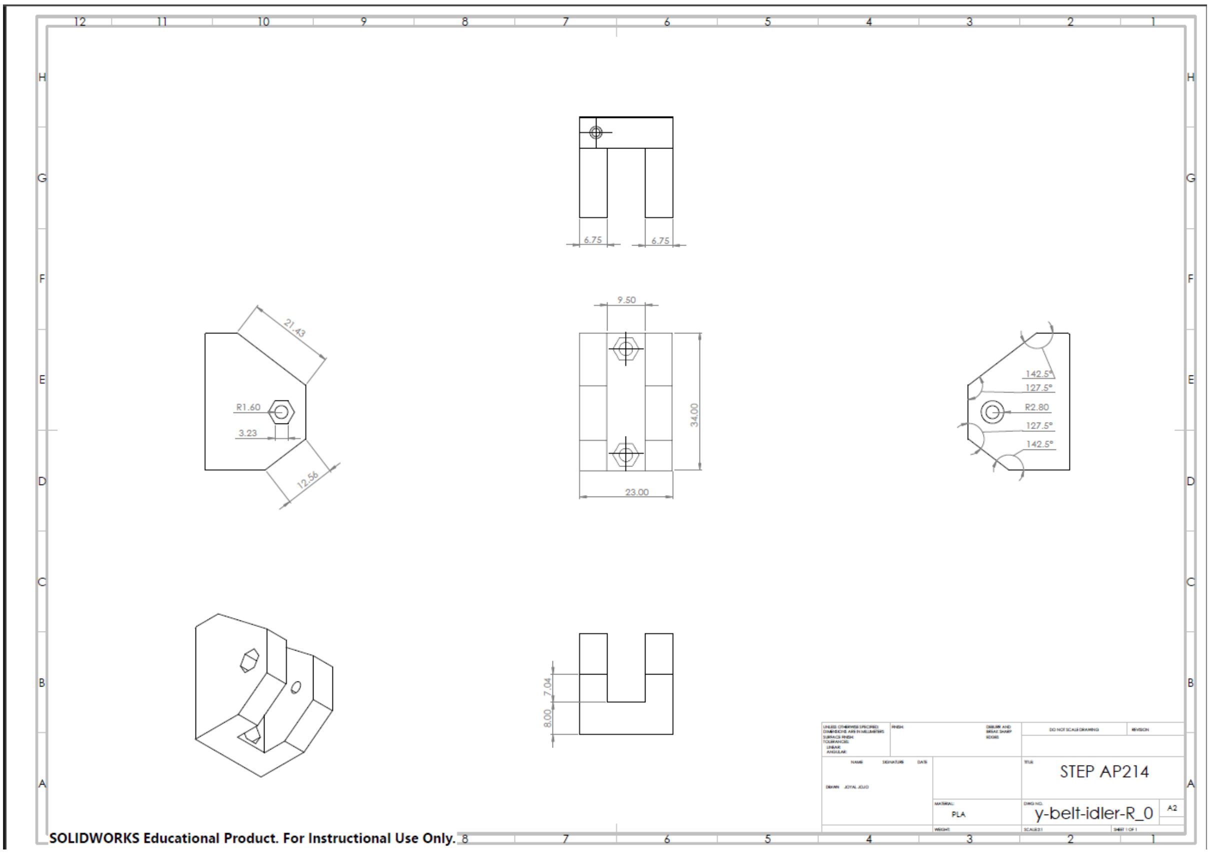
	Aluminium	Mild Steel	Stainless Steel	Titanium
Machinability	High	High	Medium	Low
Cost	Medium	Low	Medium	High
Sustainability	High	High	Medium	Low
Weight	Low	Low	Medium	High
Corrosion	Medium	High	Medium	Low

Based on Above factors Aluminium is selected.

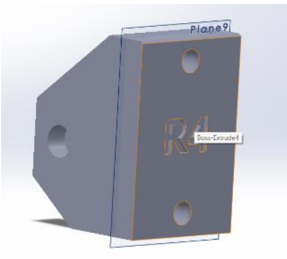
Design Simplification: CNC machining can’t perform especially undercuts or deep pockets. Simplifying the design to avoid these features can make it more suitable for 3-axis CNC machining. Tolerance and surface finish are good for CNC machining.

Tool Access and Orientation: The design should be optimized so that all features can be accessed by the cutting tools of a 3-axis CNC machine.

Cost and Time Efficiency: The design should be optimized for efficient machining to reduce cost and time per part. This could include reducing material wastage and optimizing the tool path.



- **FIRST ITERATION**



Removed unwanted Letter Labels

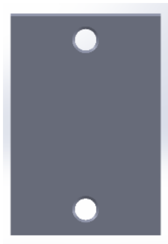
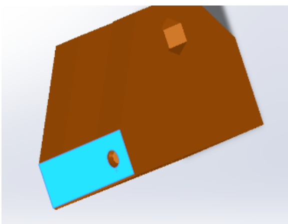


Figure 5. Part Labels on Bottom of Part.

Figure 6.



Removed Front hole.

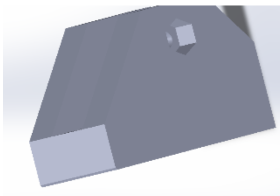


Figure 7.

Figure 8.

- **SECOND ITERATION:** Insertion of fillets

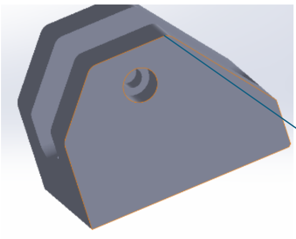


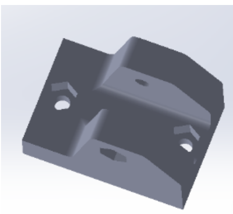
Figure 9.

Fillets

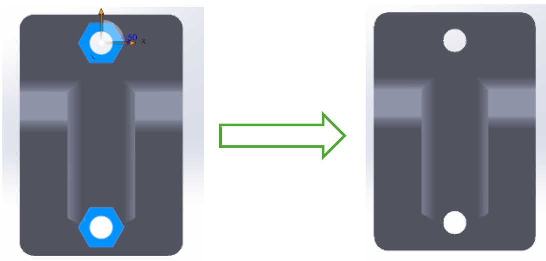
Figure 10.

- **THIRD ITERATION:** Redesign of Side walls

- **FOURTH ITERATION:** Optimisation for efficient tool path and accessibility



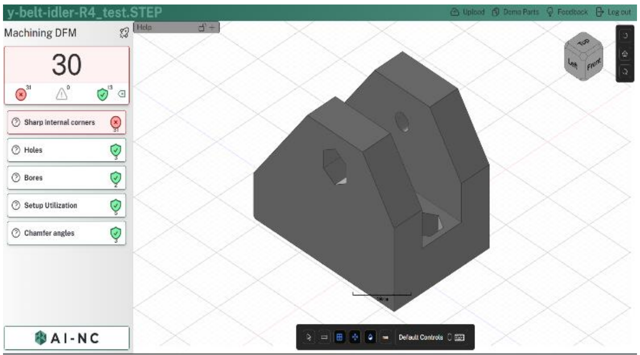
Removed Bottom fillets to save machining time.

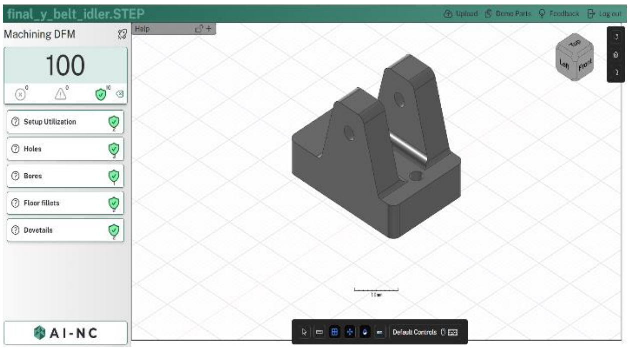


Removed Hexagonal holes(Can’t machine easily)

Location of bottom holes and top holes are constarined to change due to part functionality.

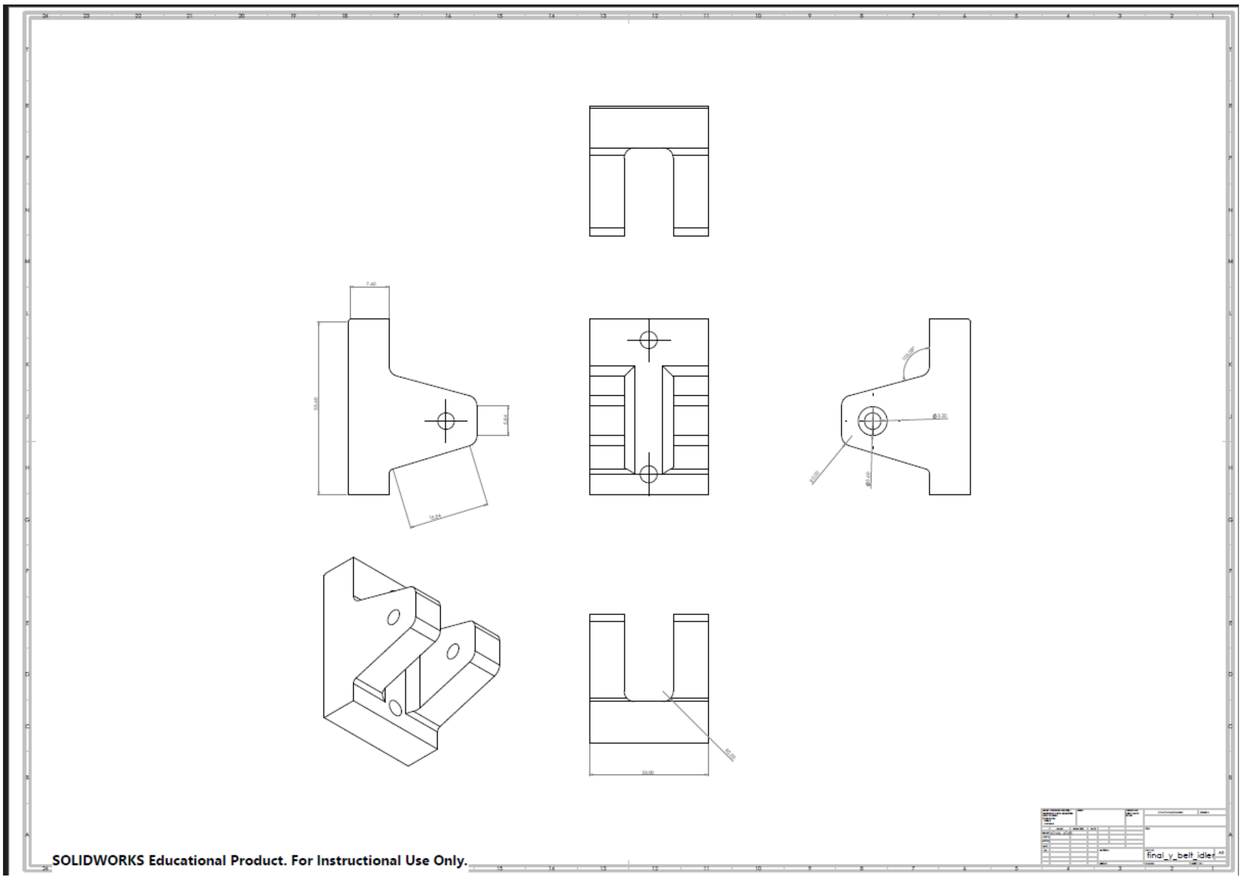
AI-NC Analysis output.





The design can be optimised further by making the side walls rectangular: - This design decrease wastage, decrease Machining time thus decreased wear and tear. Present design is having same features but increased wastage and machining time but more aesthetically pleasing. Considering assembly time both designs are having same degrees of α and β.

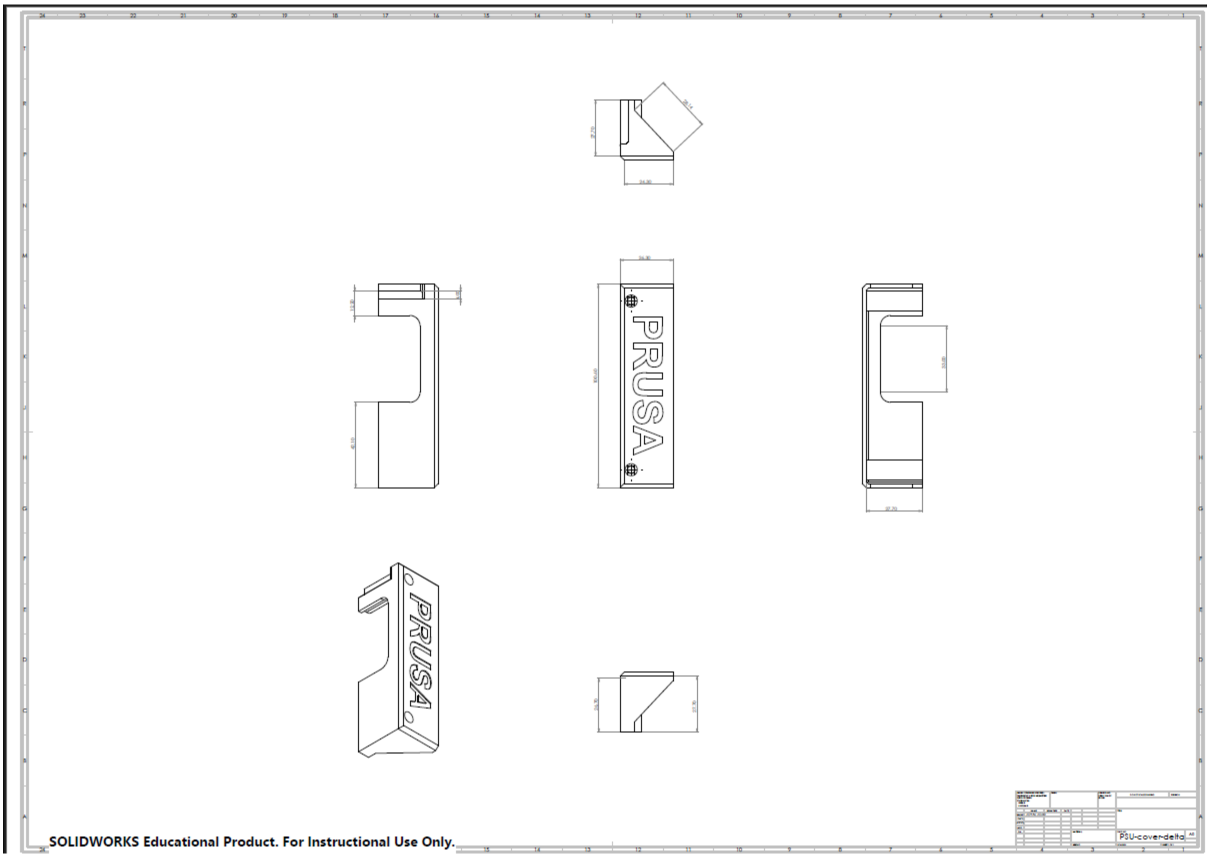
Final part Drawing



Q2: Injection Moulding Part Design

Part :908 **PSU cover**

Initial Drawing



Design Parameters:

Material Selection: we need to opt for injection moulding-grade materials that offer the desired mechanical and thermal properties for a PSU cover, such as ABS or polycarbonate.

Material	Strength	Heat Resistance	Impact Resistance	Cost	Finish Quality	Ease of Moulding
ABS	High	Good	Good	Medium	Excellent	Easy
Polycarbonate	Very High	Excellent	Very High	High	Excellent	Moderate
Polypropylene	Moderate	Good	High	Low	Good	Very Easy
Nylon (PA)	High	Excellent	High	Medium	Good to High	Moderate

Out of the materials from the above table ABS can be opted for our application.

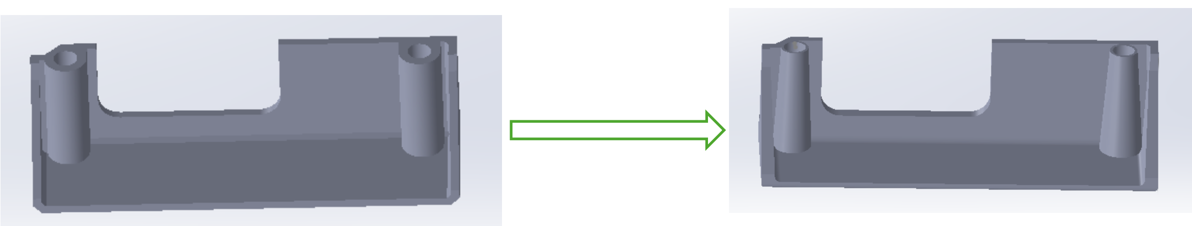
Wall Thickness: It is very important to maintain uniform wall thickness to prevent defects and ensure structural integrity without increasing material use and cycle time. 2mm wall thickness is maintained throughout the part.

Draft Angles and Radii: Incorporated draft angle of 3 mm to facilitate easy part ejection and use rounded corners with 1.5mm radius to improve material flow and part strength.

Mold Design: Avoided undercuts and complex geometries that complicate the mould and increase costs.

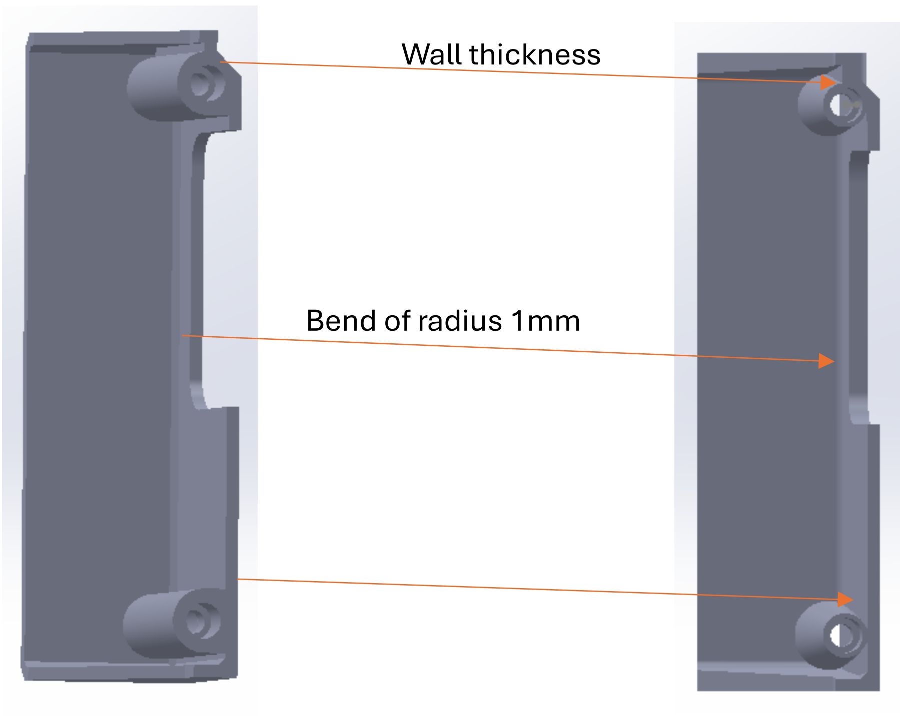
. **First Iteration:**

Draft angle for screw holes:

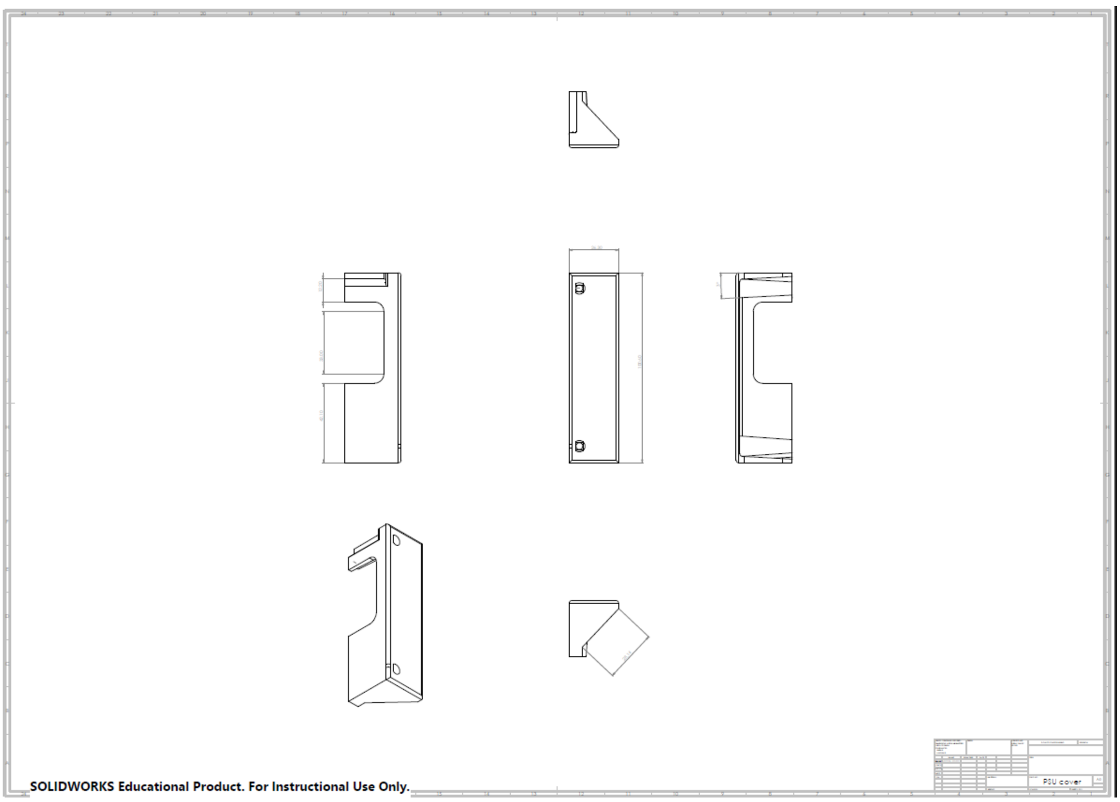


. **Second Iteration:**

Converting wall thickness to uniform everywhere(2mm) and creating bend for smooth flow of mould.



PSU Cover final drawing.



Parting line:

The parting line can be kept just above the surface. The below image shows a similar part done using injection moulding.

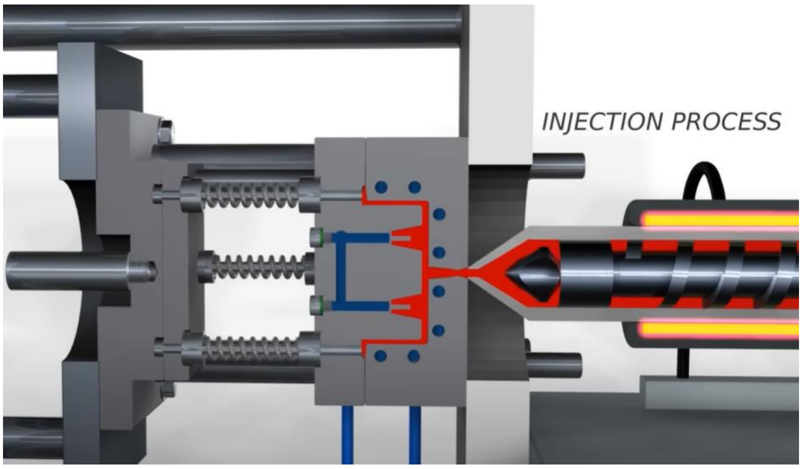
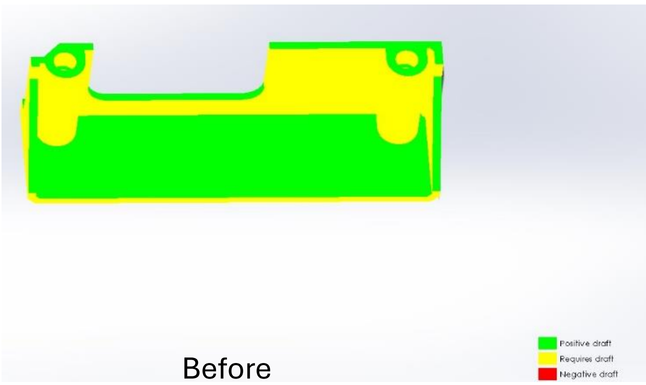


Image source: https://www.youtube.com/watch?v=b1U9W4iNDiQ

Effect of draft angle (SolidWorks draft angle analysis):

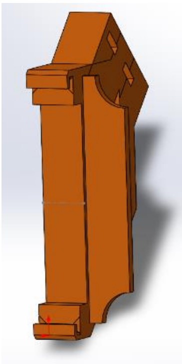


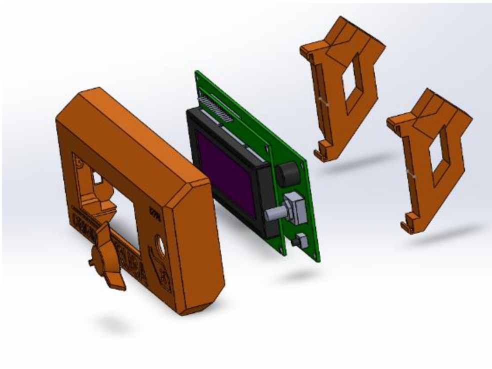
Before

After

Q3: Sheet Metal Part Design

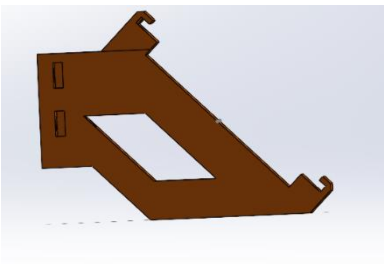
Part: 602 **LCD Support**

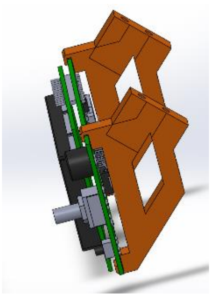


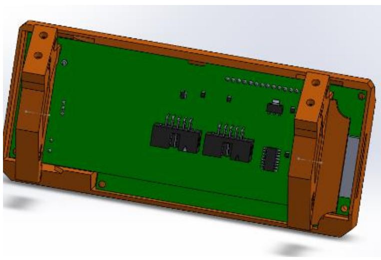


Critical constraints for converting LCD Support from 3D printed form to sheet metal.

1. Thickness of materials
2. Positioning of Holes
3. LCD Holding design.







Figures: Different views of LCD support

Material Selection: We need to choose sheet metal material that is easy to fabricate and meets the strength requirements of the part and light weight.

Material	Weight	Strength	Fabrication Ease	Corrosion Resistance	Cost	Thermal Conductivity
Aluminium	Light	Moderate	Easy	Good	Low-Mid	High
Cold Rolled Steel	Heavy	High	Moderate	Poor (needs finish)	Low	Low
Stainless Steel	Heavy	Very High	Difficult	Excellent	High	Low-Mid
Galvanized Steel	Heavy	High	Moderate	Good	Low-Mid	Low

Aluminium can be selected for our purpose.

Material Thickness: Select material thickness that provides the necessary rigidity without making the part too heavy or difficult to fabricate. Material thickness selected is 2mm by stress analysis.

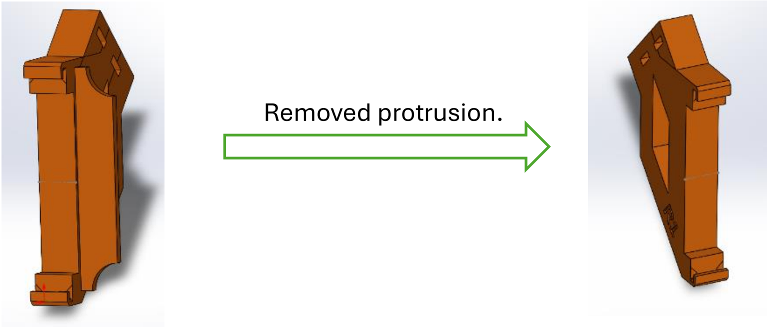
Bend Radius: Sheet metal can be bent to form various shapes, but each material has a minimum bend radius that must be observed to avoid cracking. This will affect the design of the folds and the overall shape of the part. The design selects 2.25mm bend radius with ±0.5mm tolerance.

Design Simplification: Simplify the geometry to consist of flat planes and bends. Unnecessary Holes are eliminated.

Fastening Methods: Since sheet metal parts are typically assembled with screws, rivets, or welds, we will need to incorporate features for these fastening methods. 3.5mm diameter ANSI screws are selected for this.

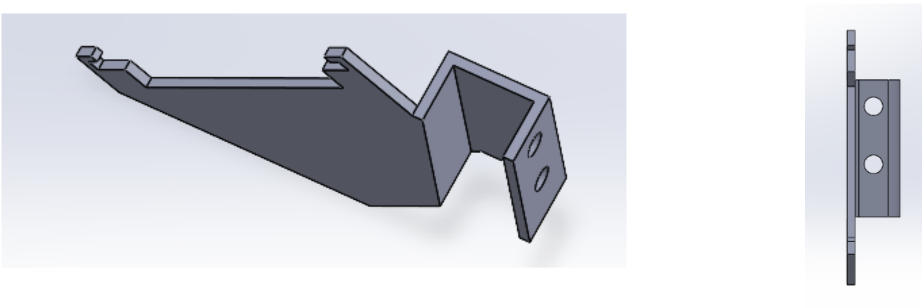
Manufacturability: Considering factors such as the ease of cutting the material, the number of bends, and the sequence of operations required to form the part. The design now consists of 3 bends and 4 holes so that it can be efficiently fabricated to keep costs down.

- **First Iteration**



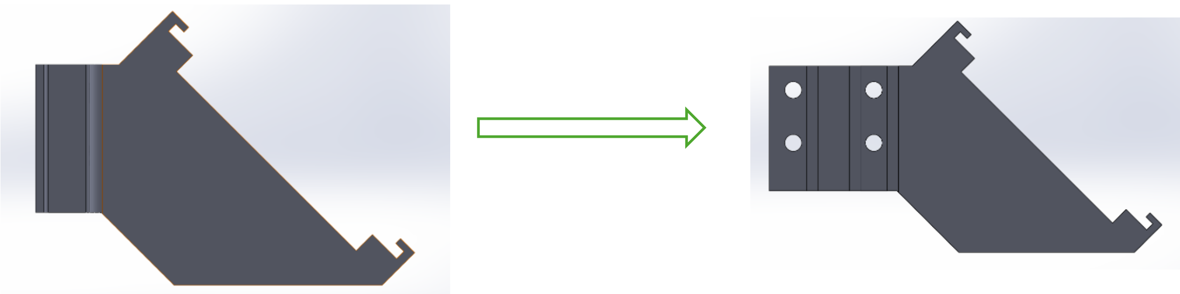
- **Second Iteration**

✓ Removed unnecessary fillets and chamfers.
✓ Reduced thickness from 9mm to 2 mm
✓ Eliminated square hole from centre.

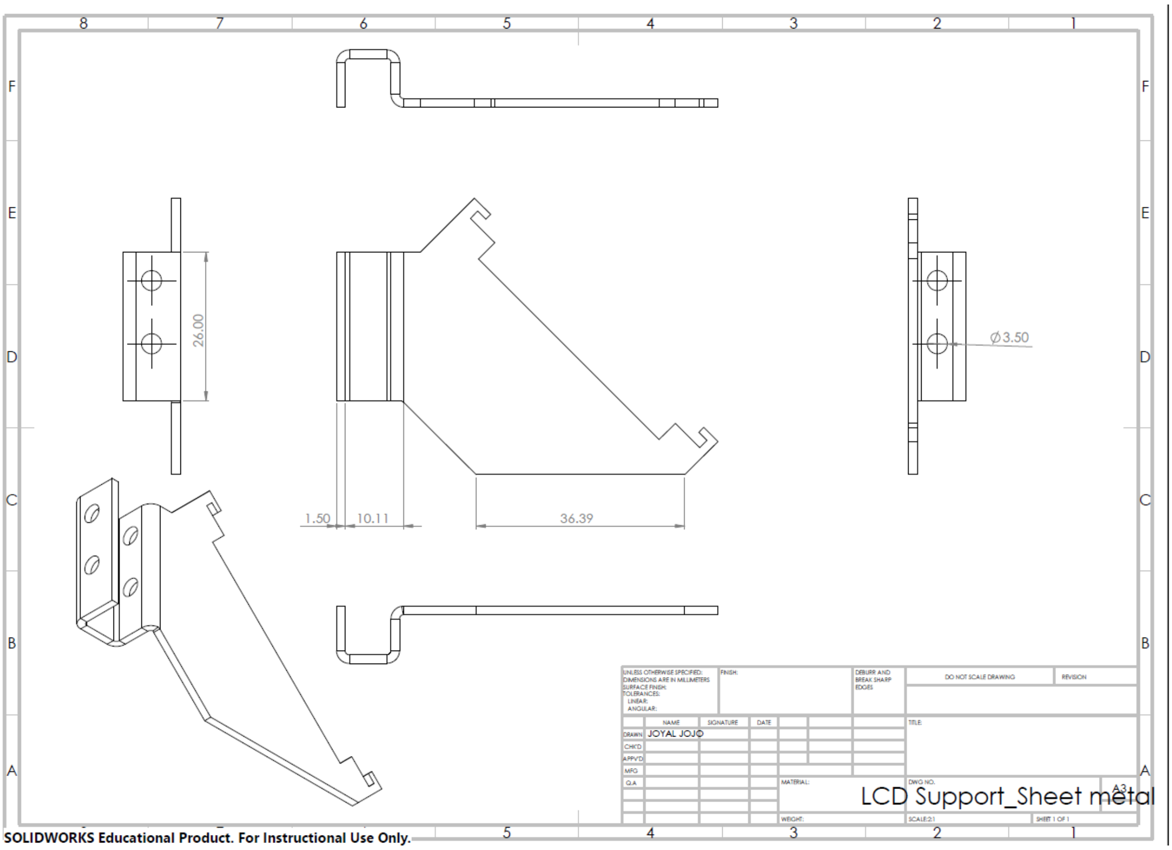


- **Third Iteration**

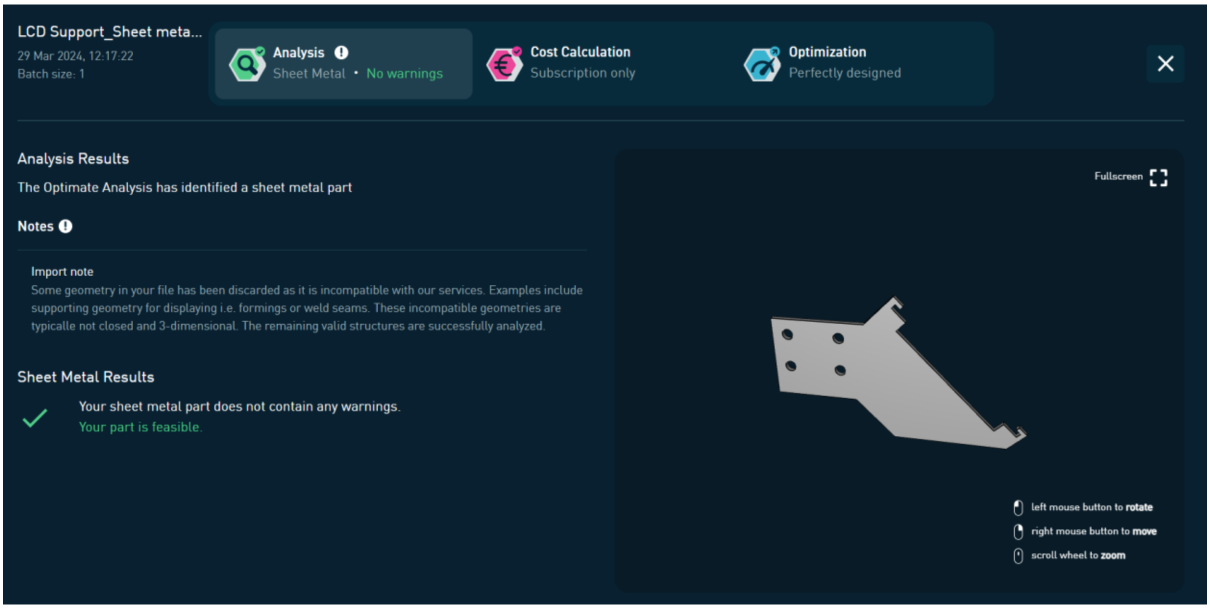
Converted the part with bends and flattened (using SolidWorks)



Redesigned part drawing



Analysis of sheet metal feasibility using Optimate



Successfully completed redesign of a non-sheet metal part to sheet metal part feasibly.